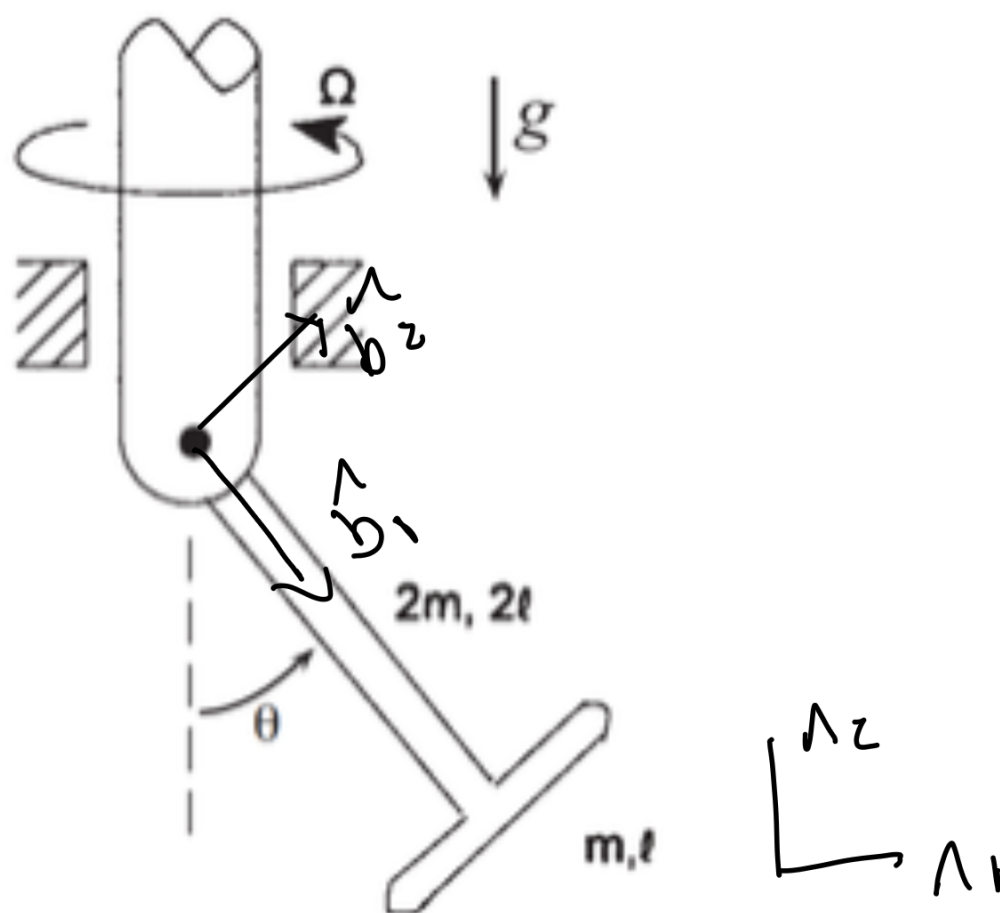


Problem 1



A bar of mass m , length ℓ is welded to a bar of mass $2m$, length 2ℓ to form the T shown. The body is pinned to a post that rotates at the constant rate Ω .

The moment of inertia tensor for the T-shaped bar is given by,

$$I = \begin{bmatrix} \frac{1}{12}m\ell^2 & 0 & 0 \\ 0 & \frac{20}{3}m\ell^2 & 0 \\ 0 & 0 & \frac{27}{4}m\ell^2 \end{bmatrix}$$

in the body frame $\{\hat{b}\}$ along the principal axes of the body, centered at the pivot point, with $\hat{b}_1$ pointing along the length- 2ℓ bar (down and to the right), $\hat{b}_2$ pointing up and to the right parallel to the length- ℓ bar, and $\hat{b}_3$ pointing out of the page.

$$\vec{\omega} = \Omega \hat{b}_2 + \dot{\theta} \hat{b}_3$$

$$= -\Omega \cos\theta \hat{b}_1 + \Omega \sin\theta \hat{b}_2 + \dot{\theta} \hat{b}_3$$

$$\vec{H} = I \vec{\omega}$$

a)

$$\begin{aligned} \vec{H} = & -\frac{\Omega}{12} mL^2 \cos\theta \hat{b}_1 \\ & + \frac{20}{3} \Omega mL^2 \sin\theta \hat{b}_2 \\ & + \frac{27}{4} mL^2 \dot{\theta} \hat{b}_3 \end{aligned}$$

$${}^N \frac{d\vec{H}}{dt} = {}^B \frac{d\vec{H}}{dt} + \omega_{B/N} \times \vec{H}$$

$$\begin{aligned} & \frac{\Omega}{12} mL^2 \sin\theta \hat{b}_1 + \frac{20}{3} \Omega mL^2 \cos\theta \hat{b}_2 \\ & + \frac{27}{4} mL^2 \ddot{\theta} \hat{b}_3 \end{aligned}$$

$\hat{b}_1$	$\hat{b}_2$	$\hat{b}_3$
$-\sin\theta$	$\sin\theta$	$\dot{\theta}$
$-\frac{\mu}{12} L^2 \cos\theta$	$\frac{20}{3} \mu L^2 \sin\theta$	$\frac{27}{4} \mu L^2 \dot{\theta}$

$$\frac{d\vec{H}}{dt} =$$

$$\left(\frac{27}{4} \mu L^2 \dot{\theta} \sin\theta - \dot{\theta} \frac{20}{3} \mu L^2 \sin\theta + \frac{\mu}{12} L^2 \sin\theta \dot{\theta} \right) \hat{b}_1$$

$$\left(-\frac{\mu}{12} L^2 \cos\theta \dot{\theta} + \frac{27}{4} \mu L^2 \sin\theta \dot{\theta} - \frac{20}{3} \mu L^2 \cos\theta \dot{\theta} \right) \hat{b}_2$$

$$\left(-\frac{20}{3} \mu L^2 \sin\theta \cos\theta \dot{\theta} + \frac{\mu}{12} L^2 \cos\theta \sin\theta \dot{\theta} + \frac{27}{4} \mu L^2 \ddot{\theta} \right) \hat{b}_3$$

b)

$$N \frac{d\vec{H}}{dt} =$$

$$\frac{1}{6} \mu L^2 \ddot{\theta} \sin \theta \hat{b}_1$$

$$+ \frac{40}{3} \mu L^2 \dot{\theta} \cos \theta \hat{b}_2$$

$$+ \left(\frac{27}{4} \mu L^2 \ddot{\theta} - \frac{7a}{12} \mu^2 L^2 \sin \theta \cos \theta \right) \hat{b}_3$$

c)

$$r_{bar1/C} = L \hat{b}_1$$

$$r_{bar2/C} = 2L \hat{b}_1$$

$$\hat{g} = -g \hat{A}_2 = g \cos \theta \hat{b}_1 - g \sin \theta \hat{b}_2$$

...

$$\vec{F}_{2M} = 2Mg \cos \theta \hat{b}_1 - 2Mg \sin \theta \hat{b}_2$$

$$\vec{F}_M = Mg \cos \theta \hat{b}_1 - Mg \sin \theta \hat{b}_2$$

$$\sum \vec{M} = \vec{r}_{1/c} \times \vec{F}_{2M} + \vec{r}_{2/c} \times \vec{F}_M$$

$$= -2L Mg \sin \theta \hat{b}_3$$

+

$$-2L Mg \sin \theta \hat{b}_3$$

$$= -4L Mg \sin \theta \hat{b}_3$$

c)

$$\sum M = -2L \cos \theta \hat{b}_1 + 2L \sin \theta \hat{b}_2 - 4Mg \sin \theta \hat{b}_3$$

d)

$$M_{B3} = \frac{d\vec{H}}{dt} \vec{b}_3$$

$$-4\mu L g \sin\theta = \frac{27}{4} \mu L^2 \ddot{\theta} - \frac{7a}{12} \omega^2 \mu L^2 \sin\theta \cos\theta$$

$$\ddot{\theta} = \frac{\frac{7a}{12} \omega^2 \mu L^2 \sin\theta \cos\theta - 4\mu L g \sin\theta}{\frac{27}{4} \mu L^2}$$

$$\ddot{\theta} = \frac{7a}{81} \omega^2 \sin\theta \cos\theta - \frac{16}{27} \frac{g}{L} \sin\theta$$

$$\sin\theta = \theta, \cos\theta = 1$$

$$\ddot{\theta} = \frac{7a}{81} \omega^2 \theta - \frac{16}{27} \frac{g}{L} \theta$$

$$\ddot{\theta} = 0 \rightarrow \frac{16}{27} \frac{g}{L} = \frac{7a}{81} \omega^2$$

$$\omega_{crit} = \sqrt{\frac{48}{7a} \frac{g}{L}}$$

c)

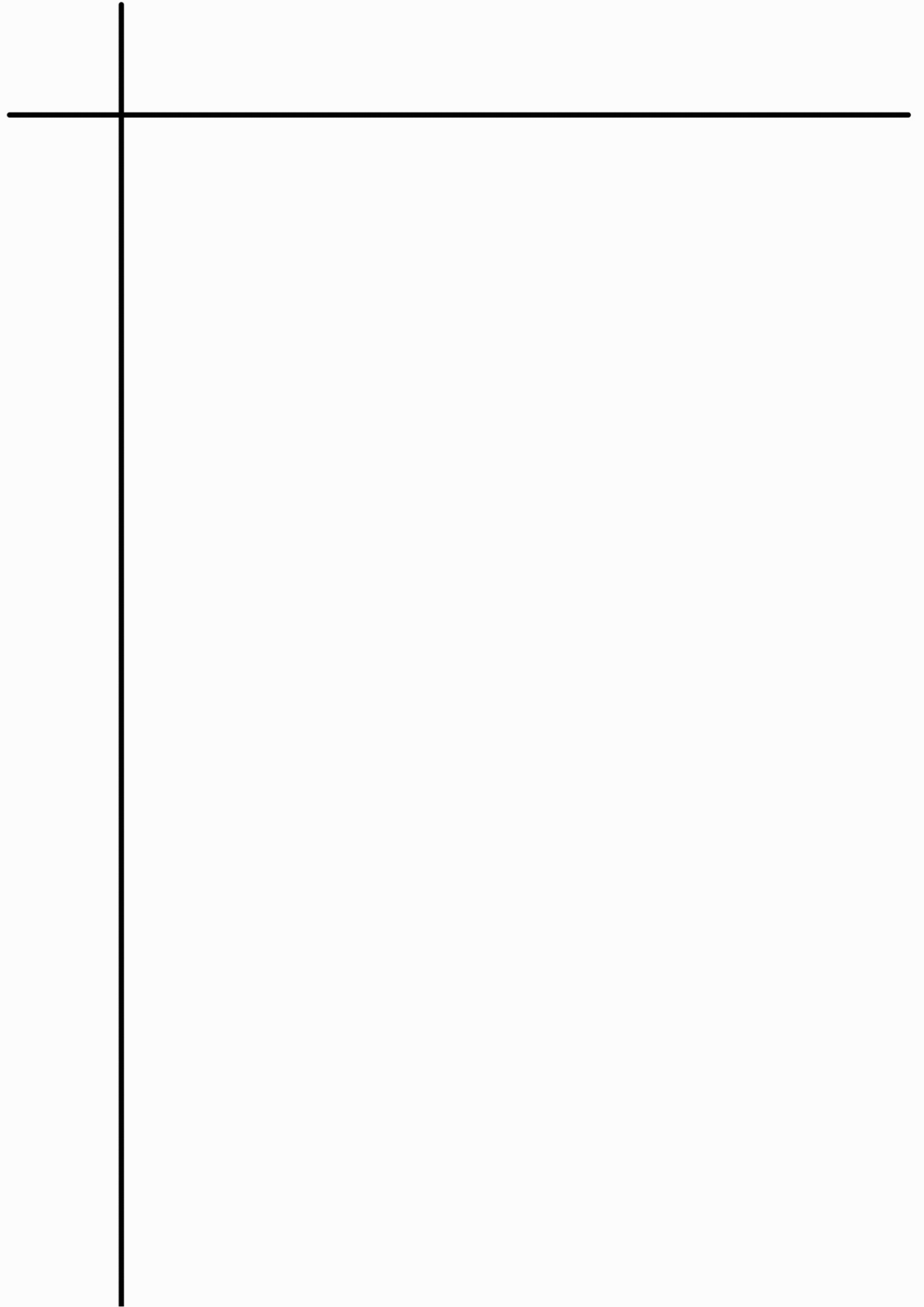
$$\dot{\theta} = v$$

$$\dot{v} = \frac{7a}{81} \Omega^2 \sin \theta \cos \theta - \frac{16}{27} \frac{g}{L} \sin \theta$$

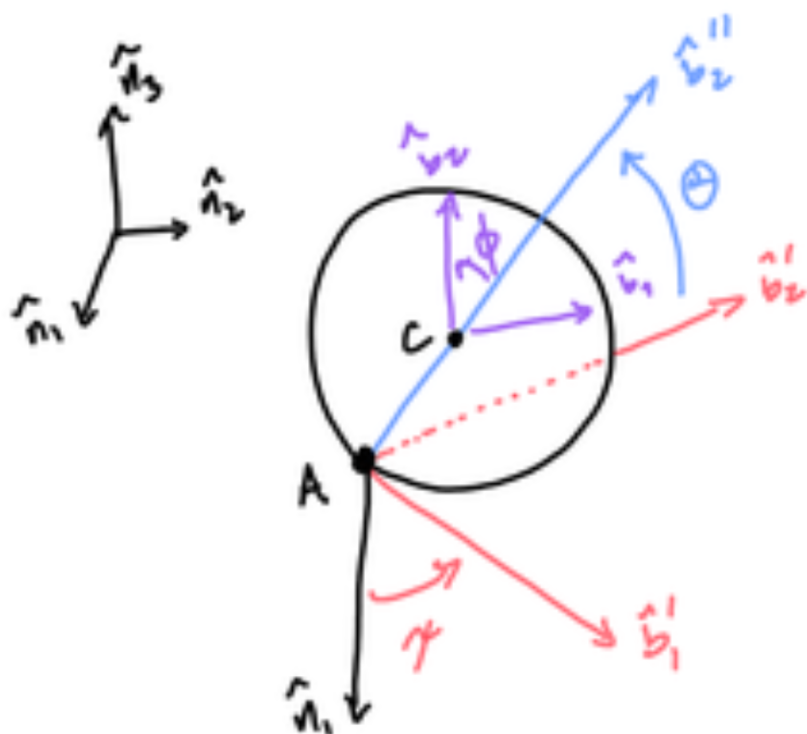
$$J = \begin{bmatrix} 0 & 1 \\ \frac{7a}{81} \Omega^2 \cos^2 \theta - \frac{7a}{81} \Omega^2 \sin^2 \theta & \end{bmatrix}$$

↓

$$J = \begin{bmatrix} 0 & 1 \\ -\frac{7a}{81} \Omega^2 - \frac{16}{27} \frac{g}{L} \cos \theta & 0 \end{bmatrix}$$



Problem 2



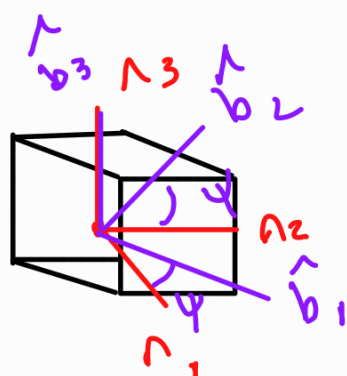
Consider a coin rolling along its edge on a flat surface. To describe its orientation, we consider the position of the center of the coin to be defined as,

$$\vec{r}_c = x\hat{n}_1 + y\hat{n}_2 + z\hat{n}_3.$$

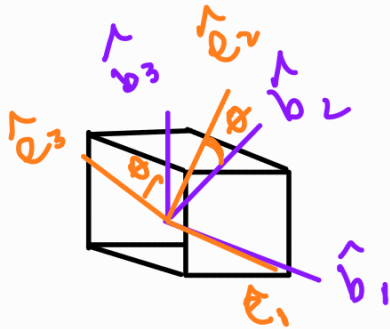
To describe its orientation, we use 3-1-3 Euler angles, just as in HW6. Consider the Euler angles to be:

First, imagine the coin to be laying flat on the ground. Rotate the heading of the coin by a counterclockwise angle ψ around the $\hat{n}_3$ axis. Second, rotate the coin upward around the $\hat{b}_1$ axis by the angle θ . If $\theta = 90^\circ$, the coin is standing on its edge, pointing in the direction of $\hat{b}_1$. Finally, we consider rolling around the $\hat{b}_3 = \hat{b}_3^*$ axis by the angle ϕ .

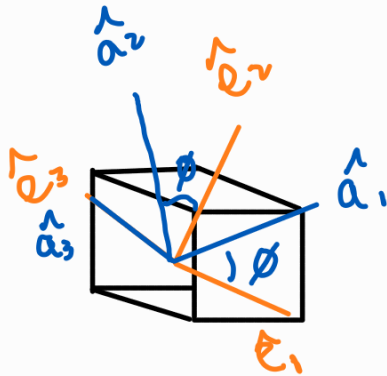
NOTE: If you're having a hard time understanding the orientation of the coin, feel free to reach out! I'll try to re-describe its orientation.



$$R(\psi) = \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



$$R(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{bmatrix}$$



$$R(\phi) = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$C: R(\phi) \cdot R(\theta) \cdot R(\psi)$$

$$\begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi \cos \theta & \cos \psi \cos \theta & \sin \theta \\ \sin \psi \sin \theta & -\cos \psi \sin \theta & \cos \theta \end{bmatrix}$$

a)

$$C = \begin{bmatrix} \cos \psi \cos \phi - \sin \psi \sin \phi \sin \theta & \sin \psi \cos \phi + \cos \psi \sin \phi \sin \theta & \sin \theta \sin \phi \\ -\cos \psi \sin \phi - \sin \psi \cos \phi \sin \theta & -\sin \psi \sin \phi + \cos \psi \cos \phi \sin \theta & \sin \theta \cos \phi \\ \sin \psi \cos \theta & -\cos \psi \sin \theta & \cos \theta \end{bmatrix}$$

$$\vec{\omega} = \dot{\psi} \hat{\Lambda}_3 + \dot{\theta} \hat{b}_1' + \dot{\phi} \hat{b}_3$$

$$\vec{\omega} = \dot{\psi} \hat{b}_3' + \dot{\theta} \hat{b}_1'' + \dot{\phi} \hat{b}_3$$

$$b' \rightarrow b'' = \underbrace{R(\phi)}_{\hat{b}_1''} R(\theta) \quad \hat{b}_2'' \quad \hat{b}_3''$$

$$\hat{b}_1' =$$

$$\hat{b}_2' =$$

$$\hat{b}_3' =$$

$$\begin{bmatrix} c\phi & s\phi c\theta & s\phi s\theta \\ -s\phi & c\phi c\theta & c\phi s\theta \\ 0 & s\theta & c\theta \end{bmatrix}$$

$$\vec{\omega} = \dot{\psi} s\theta \hat{b}_2'' + \dot{\psi} c\theta \hat{b}_3'' + \dot{\theta} \hat{b}_1'' + \dot{\phi} \hat{b}_3$$

$$b'' \rightarrow b = R(\phi)^T R(\theta) = \begin{bmatrix} c\phi & s\phi & 0 \\ -s\phi & c\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

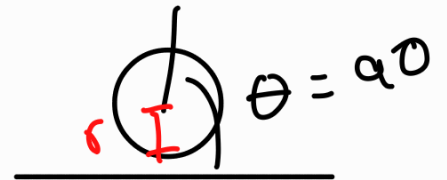
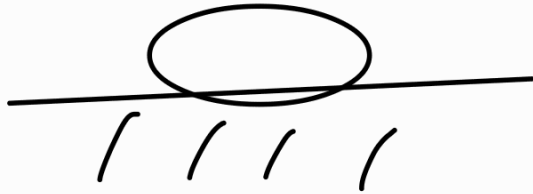
$$\vec{\omega} = -\dot{\psi} \sin\theta \sin\phi \hat{b}_1 + \dot{\psi} \sin\theta \cos\phi \hat{b}_2 \\ + \dot{\psi} \cos\theta \hat{b}_3 + \dot{\theta} \cos\phi \hat{b}_1 + \dot{\theta} \sin\phi \hat{b}_2 \\ + \dot{\phi} \hat{b}_3$$

a)

$$\vec{\omega} = (\dot{\theta} \cos\phi + \dot{\psi} \sin\theta \sin\phi) \hat{b}_1 \\ + (-\dot{\theta} \sin\phi + \dot{\psi} \sin\theta \cos\phi) \hat{b}_2 \\ + (\dot{\phi} + \dot{\psi} \cos\theta) \hat{b}_3$$

$$\alpha + \theta = 0$$

coin is flat



b)

$$\therefore z = r \sin \theta$$

$$T = \frac{1}{2} M |V|^2 + \frac{1}{2} I |\omega|^2$$

$$r = x \hat{\lambda}_1 + y \hat{\lambda}_2 + \underbrace{z \hat{\lambda}_3}_{r \tilde{\sin} \theta}$$

$$V = \dot{x} \hat{\lambda}_1 + \dot{y} \hat{\lambda}_2 + r \cos \theta \dot{\theta} \hat{\lambda}_3$$

$$T_{\text{trans}} = \frac{1}{2} M (\dot{x}^2 + \dot{y}^2 + r^2 \cos^2 \theta \dot{\theta}^2)$$

$$T_{\text{rot}} = \frac{1}{2} (I_1 \omega_1^2 + I_2 \omega_2^2 + I_3 \omega_3^2)$$

$$I_1 = I_2 = I +$$

$\therefore$

$$\frac{1}{2} I + (\omega_1^2 + \omega_2^2) + \frac{1}{2} I_a \omega_3^2$$

$$\omega_1^2 = \dot{\theta}^2 \cos^2 \theta + 2\dot{\theta} \cos \theta \dot{\psi} \sin \theta \sin \phi + \dot{\psi}^2 \sin^2 \theta \sin^2 \phi$$

$$\omega_2^2 = \dot{\theta}^2 \sin^2 \theta - 2\dot{\theta} \sin \theta \dot{\psi} \sin \theta \cos \phi + \dot{\psi}^2 \sin^2 \theta \cos^2 \phi$$

$$\omega_1^2 + \omega_2^2 = \dot{\theta}^2 + \dot{\psi}^2 \sin^2 \theta$$

$$\omega_3^2 = \dot{\phi}^2 + 2\dot{\phi} \dot{\psi} \cos \theta + \dot{\psi}^2 \cos^2 \theta$$

$$T_{\text{rot}} = \frac{1}{8} m r^2 (\dot{\theta}^2 + \dot{\psi}^2 \sin^2 \theta)$$

$$+ \frac{1}{4} m r^2 (\dot{\phi}^2 + 2\dot{\phi} \dot{\psi} \cos \theta + \dot{\psi}^2 \cos^2 \theta)$$

$$T_{trans} = \frac{1}{2} M (\dot{x}^2 + \dot{y}^2 + r^2 \cos^2 \theta \dot{\theta}^2)$$

$$T = \frac{1}{2} M \left(\dot{x}^2 + \dot{y}^2 + r^2 \dot{\theta}^2 \cos^2 \theta + \frac{1}{4} r^2 (\dot{\theta}^2 + \dot{\psi}^2 \sin^2 \theta) + \frac{1}{2} r^2 (\dot{\theta}^2 + 2 \dot{\theta} \dot{\psi} \cos \theta + \dot{\psi}^2 \sin^2 \theta) \right)$$

$$V = mg r \sin \theta$$

$\therefore$

c)

$$L = \frac{1}{2} M \left(\dot{x}^2 + \dot{y}^2 + r^2 \dot{\theta}^2 \cos^2 \theta + \frac{1}{4} r^2 (\dot{\theta}^2 + \dot{\psi}^2 \sin^2 \theta) + \frac{1}{2} r^2 (\dot{\theta}^2 + 2 \dot{\theta} \dot{\psi} \cos \theta + \dot{\psi}^2 \sin^2 \theta) \right) - mg r \sin \theta$$

$$\vec{V}_A = \vec{V}_C + \vec{\omega} \times \vec{r}_{A/C}$$

$$r_{A/C} = -r \hat{b}_z$$

b_1	b_2	b_3
ω_1	ω_2	ω_3
0	-r	0

$$\vec{\omega} \times \vec{r}_{A/C} = r \omega_3 \hat{b}_1 - \omega_1 r \hat{b}_3$$

$$= (r \dot{\theta} + r \dot{\psi} \cos \theta) \hat{b}_1$$

$$= (r \dot{\theta} \cos \theta + r \dot{\psi} \sin \theta \sin \theta) \hat{b}_3$$

Need
it in

$\hat{a}$
frame

$$\rightarrow C^T (\vec{\omega} \times \vec{r}_{A/C})$$

$$C^T = \begin{bmatrix} C\psi C\theta - S\psi C\theta S\phi & -C\psi S\theta - S\psi C\theta C\phi & S\psi S\theta \\ S\psi C\theta + C\psi C\theta S\phi & -S\psi S\theta + C\psi C\theta C\phi & -C\psi S\theta \\ S\theta S\phi & S\theta C\phi & C\theta \end{bmatrix}$$

$$\dot{x} \hat{\Lambda}_1 + \dot{y} \hat{\Lambda}_2 + \dot{z} \hat{\Lambda}_3 = -\mathcal{L}^T (\vec{\omega} \times \vec{r}_{AC})$$

$$\dot{x} = -r(\dot{\phi} + \dot{\psi} \cos \theta)(\cos \psi \cos \phi - \sin \psi \cos \theta \sin \phi) +$$

$$r(\dot{\theta} \cos \phi + \dot{\psi} \sin \theta \sin \phi) \sin \psi \sin \theta$$

$$\dot{y} = -r(\dot{\phi} + \dot{\psi} \cos \theta)(\sin \psi \cos \phi + \cos \psi \cos \theta \sin \phi) +$$

$$r(\dot{\theta} \cos \phi + \dot{\psi} \sin \theta \sin \phi) \cos \psi \sin \theta$$

$$\dot{z} = r \cos \theta \dot{\theta}$$

$$L = \frac{1}{2} m \left(\dot{x}^2 + \dot{y}^2 + r^2 \dot{\theta}^2 \right. \\ \left. + \frac{1}{4} r^2 (\dot{\theta}^2 + \dot{\psi}^2 \sin^2 \theta) \right. \\ \left. + \frac{1}{2} r^2 (\dot{\phi}^2 + 2\dot{\phi}\dot{\psi} \cos \theta + \dot{\psi}^2 \cos^2 \theta) \right)$$

$$- m g r \sin \theta$$

$$A \vec{\gamma} = b$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = \sum \lambda_i \frac{df_j}{dq_i}$$

$$x: m \ddot{x} = \lambda_1$$

$$y: m \ddot{y} = \lambda_2$$

$$\psi: I_t \ddot{\psi} s^2 \theta + \sum I_t \dot{\psi} \dot{\theta} s \theta c \theta + I_a (\ddot{\theta} + \ddot{\psi} c \theta - \dot{\psi} \dot{\theta} s \theta) c \theta - (\dot{\theta} + \dot{\psi} c \theta) \dot{\theta} s \theta =$$

$$\left(r c \theta c \psi c \theta - r s \psi s \theta \right) \lambda_1$$

$$\left(r c \theta s \psi c \theta + r c \psi s \theta \right) \lambda_2$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\psi}} \right)$$

$$\frac{\partial L}{\partial \psi} = 0$$

$$\frac{\delta f_1}{\delta \dot{\psi}}$$

$$\frac{\delta f_2}{\delta \dot{\psi}}$$



This took so long

$$\theta: (mr^2 c^2 \theta + I_e) \ddot{\theta} - mr^2 \dot{\theta}^2 s \theta c \theta - I_e \dot{\psi}^2 s \theta c \theta + I_a \dot{\psi} s \theta (\dot{\theta} + \dot{\psi} c \theta) + mgr c \theta = r s \theta c \theta (\lambda_2 c \psi - \lambda_1 s \psi)$$

$$\phi: I_a (\ddot{\phi} + \ddot{\psi} c \theta - \dot{\psi} \dot{\theta} s \theta) \quad \left. \vphantom{\ddot{\phi}} \right\} \frac{\partial \mathcal{L}}{\partial \phi} = 0$$
$$= \lambda_1 r (c \psi c \phi - s \psi c \theta s \phi) + \lambda_2 r (s \psi c \phi + c \psi c \theta s \phi)$$

11
A

1	0	0	0	0	0	0	0	0
0	1	1	0	0	0	0	0	0
0	0	0	1	0	0	0	0	0
0	0	0	0	1	0	0	0	0
0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0
0	0	0	0	0	0	0	1	0
0	0	0	0	0	0	0	0	1

$\theta_1^2 + I$
 $\theta_2^2 + I$
 $\theta_3^2 + I$
 $\theta_4^2 + I$
 $\theta_5^2 + I$
 $\theta_6^2 + I$
 $\theta_7^2 + I$
 $\theta_8^2 + I$
 $\theta_9^2 + I$

I can't properly
put A by hand
so it will be in latex

See latex for plots

